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$$\begin{aligned} I &= \int_1^2 x^2 \cosh(x) \cdot dx \\ \begin{cases} u = x^2 \\ dv = \cosh(x) \cdot dx \end{cases} &\Rightarrow \begin{cases} du = 2x \cdot dx \\ v = \sinh(x) \end{cases} \\ \Rightarrow I &= [x^2 \sinh(x)]_1^2 - 2 \int_1^2 x \sinh(x) \cdot dx \\ \begin{cases} u = x \\ dv = \sinh(x) \cdot dx \end{cases} &\Rightarrow \begin{cases} du = dx \\ v = -\cosh(x) \end{cases} \\ \Rightarrow I &= [x^2 \sinh(x)]_1^2 - 2 [x \cosh(x) - \sinh(x)]_1^2 \\ \Rightarrow I &= [x^2 \sinh(x) - 2x \cosh(x) + 2 \sinh(x)]_1^2 \\ &= 6 \sinh(2) - 4 \cosh(2) - 3 \sinh(1) + 2 \cosh(1) \end{aligned}$$

References